INCH-POUND
A-A-50606(SH)
13 September 1993

COMMERCIAL ITEM DESCRIPTION

PORTABLE HYDRAULIC ACCESS AND RESCUE SYSTEM (PHARS)
(SHIPBOARD MODELS)

1. Abstract.

1.1 The PHARS is a critical piece of equipment used for shipboard damage control and rescue that must function reliably in an extremely hostile and adverse environment. It shall be a commercially produced, commercially marketed item which has been demonstrated to function reliably in the shipboard damage control environment either through actual, historical usage in this application or through compliance with the salient characteristics and market acceptability criteria of this Commercial Item Description (CID). Where available, the components of the PHARS shall comply with the applicable industry standards as appropriate.

2. Salient Characteristics.

2.1 Physical Characteristics.

2.1.1 General. The PHARS must be a portable unit. The PHARS shall be Diesel powered and capable of operation on shipboard and at sea environment.

2.1.2 Specific.

2.1.2.1 The Primary Power Unit shall include a Diesel powered hydraulic pump and other components necessary to support operation of the tools.

2.1.2.2 The power unit shall be enclosed in a roll cage type frame with attached fuel tank, and shall not exceed 160 pounds, including fuel and hydraulic fluid, to enable it to be transported by two men.

2.1.2.3 The system shall meet performance requirements when the Diesel engine is fueled with JP-5 or F-76 fuel.

Beneficial comments (recommendations, additions, deletions) an any pertenent data which may be of use in improving this document should be addressed to: Commander, SEA 03Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy., Arlington, VA 22242-5160 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4240

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2.1.2.4 The PHARS shall have sufficient fuel capacity for at least three (3) hours of continuous operation.

2.1.2.5 Hydraulic Hoses shall have a total length of 175 feet and shall be provided in 2-50 foot and and 3-25 foot lengths. They shall be provided on one (1) or two (2) hose reels.

2.1.2.6 The hydraulic hoses shall be provided with quick disconnect couplings with lock rings.

2.1.2.7 The system hydraulic pressure shall not exceed 10,500 PSIG. The PHARS hydraulic fluid shall be in accordance with MIL-H-83282, MIL-H-22072 or MIL-H-19457C (or commercial equivalent).

2.1.2.8 A Remote Manifold Block shall be provided to enable the operator to secure pressure and change tools near the operating site without the need to position a second operator at the Primary Power Unit.

2.1.2.9 Tools shall include:

(a) One (1) Spreader with interchangeable spreading tips and cutting tips.
(b) One (1) Medium size Ram with tip packages of wedges and piercing tips.
(c) One (1) Cutter.

2.1.2.10 The PHARS shall have a "Dead Man" control system on all tools.

2.1.2.11 The spreader, cutter, and ram with attachments, shall not exceed 70 pounds each and shall be able to pass through a 24 inch circular opening.

2.1.2.12 The following spare parts shall be provided in a metal tool box:

(a) Spreader cutting blades or cutting tips (1 pair).
(b) Cutter blades (1 pair).
(c) Ram V-block (1 each).
(d) Ram Pierce Point (1 each).
(e) Special Maintenance Tools (as required) (1 set).
(f) Spare Hydraulic Seals for the Spreader, Ram and Cutter (1 set for each tool).

2.1.2.13 The PHARS shall be delivered with enough hydraulic fluid to fill the unit and make it ready for operation. The unit may be delivered either wet or dry.

2.1.2.14 All items offered to comply with this Commercial Item Description must have been sold on the commercial market or to the Government, local, state or federal.

2.2 Operational Characteristics.

2.2.1 General. The PHARS is a hydraulic power tool for use in shipboard damage control and rescue. It shall be capable of successful operation against typical ship interior materials and spaces in order to fulfill the damage control/rescue mission.

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2.2.2 Specific.

2.2.2.1 Rough Usage. The PHARS power unit, hose reel, spreader, cutter and ram shall each be able to withstand a fall and operate as required after being raised two (2) feet above a steel plate simulating a ships deck. The two (2) feet shall be measured from the lowest point of the assembly when raised.

2.2.2.2 Performance. The PHARS shall meet the performance requirements when in operation using the to-be-supplied Diesel power unit, a total of 175 feet of hydraulic hose connected between the power unit and the tool in use and with the remote manifold block connected to the tool (including any permanently attached hydraulic hose) by not more than 25 feet of hydraulic hose.

(a) The remote manifold block shall secure the pressure to the tool to enable tools to be changed without the need to have a second operator stationed at the power unit to manipulate manifold block at the power unit.
(b) The PHARS, with tools of the operator's choice, shall be capable of making, in less than 20 minutes (including time for repairs or adjustments), a 25 inch clear opening in a 4 foot by 4 foot by 1/4 inch thick, ASTM A 36, mild steel plate vertically fixed to a suitable foundation with bracing structure for a ram fixed at a distance of not less than 28 inches from the test panel. "Clear" indicates a totally unobstructed opening.
(c) The cutter shall be capable of shearing 10 portions of steel off a length of 1-inch diameter solid ASTM A 36 steel bar stock in 15 minutes or less (including time for repairs and adjustments).
(d) The system shall be able to satisfy the performance requirements with the primary power unit inclined 15 degrees from the horizontal in either the longitudinal or transverse axis.

2.2.2.3 Force.

(a) The PHARS shall be able to meet the force requirements with the tools powered by the diesel power unit with a total of 175 feet of hydraulic hose connected between the diesel power unit and the spreader and ram. Operating the spreader and the ram from the fully closed position to the required open distance and returning the spreader and cutter to the fully closed position shall constitute one (1) complete cycle.
(b) The spreader shall be capable of generating a spread force of 14,500 pounds or greater and while returning to the fully closed position the spreader shall be capable of generating a pulling force of 7,000 pounds or greater with the force measured at the point at which pulling shackles are normally attached to the spreader and with the spreader arms cycled open to a distance of 12 inches or greater (measured at the shackle attachment point) from the fully closed position.
(c) The ram shall be capable of generating a push force of 15,000 pounds or greater when cycling open to a distance of 12 inches or greater from the fully closed position; and while returning to the fully closed position, the ram shall be capable of generating a pull force of 6,000 pounds or greater.

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2.2.2.4 The PHARS shall satisfy the operational characteristics without:

(a) Breakage of any component or leakage of any hydraulic fluid when dropped.
(b) Failure of any PHARS component to operate as specified under the operational characteristics section of the salient characteristics description.
(c) A mechanical failure of the diesel power unit, spreader, ram or cutter (excluding removable tool tips) which requires a repair or adjustment requiring more than two (2) minutes to complete or a failure that totally disables the subject component such that a complete replacement of the subject component would be required.
(d) Any hydraulic fluid leakage from the diesel power unit, the hydraulic hose, the spreader, the ram or the cutter which requires a repair or adjustment requiring more than two (2) minutes to complete or a hydraulic leak which requires a total replacement of the subject component.

2.3 Technical Manuals.

2.3.1 Technical manuals and operational documentation as currently provided to commercial customers shall be enclosed with each deliverable PHARS unit with any addendums or attachments necessary to meet the following requirements:

2.3.1.1 General. The manual shall precisely reflect the hardware configuration of the equipment/system. The amount of data required shall be determined by the complexity/simplicity of the equipment. The manual shall consist of all data (volumes, folders, inserts, specification sheets, and other documents) required for operation and maintenance of the equipment/system as determined by the contracting activity.

2.3.1.2 Safety. The manual shall contain appropriate warnings, cautions and notes immediately preceding the point in the procedure where a hazard is likely to be encountered.

2.3.1.3 Hazardous substances. Information concerning the use, transportation, handling, storage or disposal of fuel, toxic or hazardous substance shall be clearly stated and shall be consistent with Federal Protection Standards.

2.3.1.4 Illustrations. The manual shall contain illustrations to support the text. The illustrations shall be integrated with the text or their location shall be referenced in the text. Illustrations shall be suitable for locating and identifying all components significant to operation and maintenance. Photographs shall be prescreened, sharp and with good contrast. Free hand sketches shall not be acceptable.

2.3.1.5 Preparation for use. The manual shall contain instructions for unpacking, assembling and readying for operation.

2.3.1.6 Operation. Operating instructions shall include the principles of operation; illustrations and explanations of the uses and functions of all controls and indicators; initial set up; initial settings; and start up, normal operation, shutdown and emergency shut down procedures.

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2.3.1.7 Maintenance. The manual shall reflect the manufacturer's maintenance philosophy for the hardware and shall be delineated such that it can be applied by the intended Government user. The manual shall include preventive and corrective maintenance and overhaul procedures with lists of required test equipment, special tools and materials. The nomenclature, part/model number and manufacturer of special tools, equipment, cleaning agents, lubricants and hydraulic fluids shall be listed. Diagnostic tests and tests to ensure satisfactory performance shall be detailed.

2.3.1.8 Storage. The manual shall contain general requirements for storage and any special precautions that must be taken for shipboard storage. Temperature, ventilation, covering, preservation and periodic operation requirements shall be detailed.

2.3.1.9 Disposal. Unique disposal requirements shall be delineated.

2.3.1.10 Parts list, illustrations and diagrams. The manual shall include a parts list containing positive identification of parts for support of the equipment and shall include the actual manufacturer/vendor and their part number or generic description necessary to obtain replacement parts. Clear and legible illustrations shall identify component parts and parts relationships. System diagrams shall be included as necessary.

2.4 Provisioning Data. Provisioning data shall be provided in the contractor's format. All lines, symbols, letters and numerals shall be readable.

2.4.1 Interim Provisioning Data.

2.4.1.1 Interim Support Items List. The interim support items list shall contain those repair parts required between the first delivery of the equipment and the point in time that long term provisioning has been accomplished. The interim support items list shall include piece part information on all individual parts the manufacturer recommends as the most critical maintenance parts to the contracted equipment. In addition, the items listed shall include any part requiring maintenance replacement during the first two years of operation and any accessories or attachments. CAGE and part number information should be for the actual manufacturer of the complete finished part as used in the configuration being delivered under this acquisition. Data required for each item is as follows:

Manufacturer's CAGE
Manufacturer's Part Number
Nomenclature
Unit Price
Quantity Installed Per End Item
Existing NSN (if known)
Remarks (mandatory change-out or other part information)

2.4.1.2 Supplementary Provisioning Technical Documentation. Supplementary provisioning technical documentation shall be provided for those items on the interim support items list. Supplementary provisioning technical documentation is technical data used to describe parts of equipment. It consists of data such as specifications, standards, drawings, photographs, sketches, descriptions, assembly

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and general arrangement drawings, schematic diagrams, wiring and cabling diagrams needed to indicate the physical characteristics, actions and function of the item.

(a) Supplementary provisioning technical documentation shall be provided in the following order of precedence, as available:

- Government or recognized industry specifications or standards: identification only, copies are not required.

- Commercial drawings. No effort is to be expended by the contractor to change existing commercial drawings.

- Commercial catalogs or catalog description.

- Sketches or photographs with brief descriptions of dimensional, material, mechanical, electrical, or other descriptive characteristics. When sketches or photographs are provided for an assembly, a parts list shall also be provided.

2.4.2 Full Provisioning Technical Documentation. The full provisioning parts list and supplemental provisioning parts list will be a top down breakdown of the equipment. Information will be provided as required for the interim support items list and its supplemental provisioning technical data except that parts information provided will be for all parts in the equipment provided under solicitation and supplemental provisioning technical data will be provided for any item on the full provisioning parts list not previously identified in the interim support items list.

3. Quality Assurance Provisions.

3.1 Contractor Certification Statement.

3.1.1 The contractor shall certify, and maintain objective quality evidence, that the product offered meets this Commercial Item Description and its salient characteristics, and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices. Further, where applicable, the contractor shall maintain quality evidence that all components, parts, materials, conform to the appropriate industry standard for that item, such as (but not limited to): hydraulic hoses, materials, engines, pumps, hydraulic tools. Proof of such conformance shall be provided with the contractor's proposal, prior to first article and thereafter as may be otherwise provided for under the provisions of the contract.

3.1.2 If required, the proposal shall contain the results of tests and measurements performed to demonstrate conformance with the salient characteristics. Tests and measurements shall be performed by an independent laboratory and shall be wittnessed by the Government. Only one (1) unassisted manufacturer's representative operator shall operate the spreader, ram and cutter. The same ram, cutter and spreader shall be used for all tests. Breakage and change out of only one (1) set of ram tips, spreader tips and cutter blades are allowed. Test time limits shall not be extended to facilitate tip and blade changes. No saws, drills or torches shall be used to assist the PHARS tools or attachments. The report shall include a copy of the laboratory analysis of the steel used during the tests.

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3.2 Market Acceptability Criteria.

3.2.1 The PHARS is used in life threatening, life endangering situations in an extremely hostile and adverse environment. As such its performance and reliability in this usage cannot be compromised. The following criteria will be used to evaluate the product described by this Commercial Item Description.

3.2.2 The company offering the item must have been producing the model being tendered under the solicitation or generically similar models for at least 3 years.

3.2.3 The company must have produced at least 1000 generically similar or identical models to that being tendered.

3.2.4 The company must have produced at least 100 of the model (s) being tendered with the exception of the Diesel engine power unit. The diesel engine must be a standard commercial model, fueled with JP-5 or F-76 fuel, and the remaining operating components of the power unit must have been produced at least 100 times in that configuration or combination.

3.2.5 The item offered must have been sold to the commercial market or to the Government, and the contractor shall certify and provide supporting documentation as to its serviceability and the long term (five (5) years) availability of all necessary maintenance and logistics support parts and materials.

3.3 Warranty. The item offered shall include the standard warranty given to the commercial market beginning with the date of delivery of the individual item.

4. Regulatory Requirements. All appropriate regulatory requirements that apply to items offered on the commercial market shall apply to those offered to the Government.

5. Packaging and Marking.

5.1 Packaging. Unless specific, unique destinations require special packaging to be defined by the contract, the following packaging requirements apply:

5.1.1 The PHARS shall be packed in accordance with the contractors normal method for commercial market distribution using shipping containers that comply with Uniform Freight Classification or National Motor Freight Classification rules.

5.2 Marking. The following marking requirements apply:

Name of Supplier
Point of Origin
Contents by Model/Identification number and quantity
Contract number
Name of Customer
Point of Destination
Point of Contact if applicable.

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6. Notes.

6.1 The acquisition agency should specify the preferred options permitted herein and should include the following information in the acquisition documents:

1. Title and date of this CID.
2. Requirements for unique levels of packing, if required.
3. Requirements for unique/additional marking, if required.
4. When this CID is used for acquisition the commercial item certification clause, DFARS 252.211-7012, must appear in the solicitation.

6.2 National Motor Freight Classification is available from the American Trucking Association, ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.

6.3 Uniform Freight Classification is available from the Uniform Classification Committee; Attn: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.

Preparing activity:
Navy - SH
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-50606(SH)	93/09/13

3. DOCUMENT TITLE
Portable Hydraulic Access and Rescue System (PHARS) (Shipboard Models)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL POINT OF CONTACT (TOPC): Mr. C.T. Loeser (05G1) PLEASE ADDRESS ALL CORESPONDENSE TO:	b. TELEPHONE (Include Area Code) (1) Commercial: (703)602-9295 (2) AUTOVON: 8-332-9295
c. ADDRESS (Include Zip Code) Commander, SEA 03Q42 Naval Sea Systems Command 2531 Jefferson Davis Hwy Arlington, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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